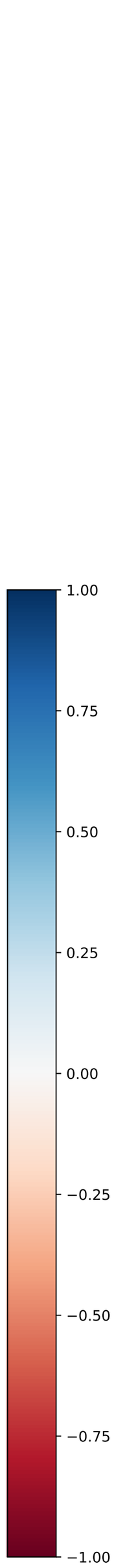
window R0 (days 0-12)

	Persistence	Open loop (info-matched)	Open loop (structure-rich)*	LSTM (v1.0)	LSTM	PINN, single-stage	CL-PINN	PINN-θ	EKF	EKS	IEKS	Aug. EKF	Aug. EKS	Online EKF*
K0 Ie A0	0.00	0.67	1.00	-0.74	n/a	0.55	0.58	0.32†	0.08	-0.14	-0.48	-0.18	-0.34	0.08
K0 Ie As	0.00	0.67	1.00	n/a	n/a	n/a	n/a	n/a	0.01	-0.25	n/a	-0.83	-0.80	0.01
K0 Ie Al1	0.00	0.73	0.97	n/a	n/a	n/a	n/a	n/a	0.20	-0.01	n/a	-0.41	-0.44	0.20
K0 Ie Al2	0.00	0.71	0.95	n/a	n/a	n/a	n/a	n/a	0.19	-0.01	n/a	-0.44	-0.42	0.19
K0 Ic A0	0.00	0.53	0.62	n/a	n/a	n/a	0.44	n/a	0.25	0.18	-0.16	-0.73	-1.02	0.25
K0 Ic As	0.00	0.53	0.62	n/a	n/a	n/a	n/a	n/a	0.20	0.11	n/a	-1.42	-1.48	0.20
K0 Ic Al1	0.00	0.62	0.68	n/a	n/a	n/a	n/a	n/a	0.36	0.30	n/a	-0.85	-0.96	0.36
K0 Ic Al2	0.00	0.62	0.68	n/a	n/a	n/a	n/a	n/a	0.36	0.30	n/a	-0.91	-0.99	0.36
K0 Ib A0	0.00	0.53	0.63	n/a	n/a	n/a	n/a	n/a	0.27	0.25	n/a	0.04	0.06	0.27
K0 Ib As	0.00	0.53	0.63	n/a	n/a	n/a	n/a	n/a	0.23	0.18	n/a	-0.34	-0.22	0.23
K0 Ib Al1	0.00	0.62	0.68	n/a	n/a	n/a	n/a	n/a	0.38	0.35	n/a	-0.06	0.02	0.38
K0 Ib Al2	0.00	0.61	0.69	n/a	n/a	n/a	n/a	n/a	0.38	0.35	n/a	-0.11	-0.02	0.38
K0 off-steady Ie A0	0.00	0.73	1.00	n/a	n/a	n/a	n/a	n/a	0.26	0.08	n/a	-0.05	-0.14	0.26
K0 off-steady Ie As	0.00	0.69	0.76	n/a	n/a	n/a	n/a	n/a	0.19	-0.01	n/a	-0.54	-0.57	0.19
K0 off-steady Ie Al1	0.00	0.69	0.81	n/a	n/a	n/a	n/a	n/a	0.37	0.24	n/a	-0.21	-0.31	0.37
K.25 Ie A0	0.00	0.17	0.48	n/a	n/a	n/a	n/a	n/a	-0.24	-0.23	n/a	-0.10	-0.29	-0.24
K.25 Ie As	0.00	0.20	0.50	n/a	n/a	n/a	n/a	n/a	-0.11	-0.03	n/a	-0.70	-0.68	-0.11
K.25 Ie Al1	0.00	0.26	0.50	n/a	n/a	n/a	n/a	n/a	-0.04	0.04	n/a	-0.54	-0.54	-0.04
K.25 Ie Al2	0.00	0.26	0.50	n/a	n/a	n/a	n/a	n/a	-0.06	0.02	n/a	-0.62	-0.60	-0.06
K.25 Ic A0	0.00	0.14	0.32	n/a	n/a	n/a	n/a	n/a	-0.33	-0.40	n/a	-0.36	-0.64	-0.33
K.25 Ic As	0.00	0.18	0.36	n/a	n/a	n/a	n/a	n/a	-0.18	-0.14	n/a	-1.05	-1.14	-0.18
K.25 Ic Al1	0.00	0.24	0.38	n/a	n/a	n/a	n/a	n/a	-0.10	-0.05	n/a	-0.79	-0.90	-0.10
K.25 Ic Al2	0.00	0.25	0.38	n/a	n/a	n/a	n/a	n/a	-0.13	-0.08	n/a	-0.85	-0.94	-0.13
K.25 Ib A0	0.00	0.14	0.32	n/a	n/a	n/a	n/a	n/a	-1.31	-1.84	n/a	-0.94	-1.37	-1.31
K.25 Ib As	0.00	0.18	0.36	n/a	n/a	n/a	n/a	n/a	-0.10	-0.04	n/a	-0.07	0.03	-0.10
K.25 Ib Al1	0.00	0.24	0.38	n/a	n/a	n/a	n/a	n/a	-0.03	0.05	n/a	0.01	0.11	-0.03
K.25 Ib Al2	0.00	0.25	0.38	n/a	n/a	n/a	n/a	n/a	-0.05	0.03	n/a	-0.03	0.09	-0.05
K.5 Ie A0	0.00	-0.39	0.03	n/a	n/a	n/a	-1.08	n/a	-1.12	-1.05	-1.28	-0.04	-0.24	-1.12
K.5 Ie As	0.00	-0.03	0.32	n/a	n/a	n/a	n/a	n/a	-0.59	-0.54	n/a	-0.56	-0.59	-0.59
K.5 Ie Al1	0.00	-0.03	0.25	n/a	n/a	n/a	n/a	n/a	-0.60	-0.55	n/a	-0.52	-0.56	-0.60
K.5 Ie Al2	0.00	-0.04	0.24	n/a	n/a	n/a	n/a	n/a	-0.68	-0.63	n/a	-0.59	-0.63	-0.68
K.5 Ic A0	0.00	-0.40	-0.08	n/a	n/a	n/a	-1.37	n/a	-1.78	-2.22	-2.86	-1.10	-1.55	-1.78
K.5 Ic As	0.00	-0.04	0.23	n/a	-0.16	n/a	-0.56	n/a	-0.59	-0.50	-0.54	-0.42	-0.46	-0.59
K.5 Ic Al1	0.00	-0.03	0.18	n/a	n/a	n/a	-0.65	n/a	-0.60	-0.51	-0.53	-0.35	-0.40	-0.60
K.5 Ic Al2	0.00	-0.04	0.16	n/a	n/a	n/a	n/a	n/a	-0.67	-0.58	n/a	-0.39	-0.44	-0.67
K.5 Ib A0	0.00	-0.40	-0.08	n/a	n/a	n/a	n/a	n/a	-1.88	-2.47	n/a	-0.84	-1.02	-1.88
K.5 Ib As	0.00	-0.03	0.23	n/a	n/a	n/a	n/a	n/a	-0.85	-1.18	n/a	-0.62	-1.01	-0.85
K.5 Ib Al1	0.00	-0.02	0.18	n/a	n/a	n/a	n/a	n/a	-0.84	-1.14	n/a	-0.58	-0.96	-0.84
K.5 Ib Al2	0.00	-0.04	0.17	n/a	n/a	n/a	n/a	n/a	-0.91	-1.22	n/a	-0.56	-0.94	-0.91
K.75 Ie A0	0.00	-0.92	-0.35	n/a	n/a	n/a	n/a	n/a	-1.83	-1.85	n/a	0.20	0.19	-1.83
K.75 Ie As	0.00	-0.16	0.24	n/a	n/a	n/a	n/a	n/a	-0.86	-0.89	n/a	0.02	-0.00	-0.86
K.75 Ie Al1	0.00	-0.20	0.12	n/a	n/a	n/a	n/a	n/a	-0.95	-0.97	n/a	0.03	-0.02	-0.95
K.75 Ie Al2	0.00	-0.22	0.10	n/a	n/a	n/a	n/a	n/a	-1.05	-1.07	n/a	0.01	-0.05	-1.05
K.75 Ic A0	0.00	-0.92	-0.45	n/a	n/a	n/a	n/a	n/a	-2.28	-2.63	n/a	-0.88	-1.19	-2.28
K.75 Ic As	0.00	-0.17	0.16	n/a	n/a	n/a	n/a	n/a	-0.89	-0.90	n/a	-0.54	-0.84	-0.89
K.75 Ic Al1	0.00	-0.20	0.06	n/a	n/a	n/a	n/a	n/a	-0.99	-0.96	n/a	-0.53	-0.84	-0.99
K.75 Ic Al2	0.00	-0.23	0.04	n/a	n/a	n/a	n/a	n/a	-1.09	-1.06	n/a	-0.52	-0.84	-1.09
K.75 Ib A0	0.00	-0.91	-0.44	n/a	n/a	n/a	n/a	n/a	-2.05	-2.54	n/a	-0.97	-1.07	-2.05
K.75 Ib As	0.00	-0.16	0.17	n/a	n/a	n/a	n/a	n/a	-0.75	-0.83	n/a	-0.16	-0.40	-0.75
K.75 Ib Al1	0.00	-0.19	0.07	n/a	n/a	n/a	n/a	n/a	-0.84	-0.88	n/a	-0.21	-0.40	-0.84
K.75 Ib Al2	0.00	-0.22	0.05	n/a	n/a	n/a	n/a	n/a	-0.94	-0.97	n/a	-0.18	-0.35	-0.94
K1 Ie A0	0.00	-1.40	-0.69	n/a	n/a	n/a	-2.39	-0.82	-2.46	-2.50	-2.24	0.13	0.22	-2.46
K1 Ie As	0.00	-0.24	0.20	n/a	n/a	n/a	n/a	n/a	-1.16	-1.19	n/a	0.12	0.12	-1.16
K1 Ie Al1	0.00	-0.30	0.05	n/a	n/a	n/a	n/a	n/a	-1.29	-1.32	n/a	0.14	0.10	-1.29
K1 Ie Al2	0.00	-0.33	0.02	n/a	n/a	n/a	n/a	n/a	-1.40	-1.43	n/a	0.15	0.10	-1.40
K1 Ic A0	0.00	-1.41	-0.79	n/a	n/a	n/a	n/a	n/a	-2.71	-2.56	n/a	-0.65	-0.61	-2.71
K1 Ic As	0.00	-0.26	0.12	n/a	n/a	n/a	n/a	n/a	-0.97	-0.86	n/a	-0.04	-0.25	-0.97
K1 Ic Al1	0.00	-0.31	-0.01	n/a	n/a	n/a	n/a	n/a	-1.13	-1.01	n/a	0.04	-0.08	-1.13
K1 Ic Al2	0.00	-0.34	-0.03	n/a	n/a	n/a	n/a	n/a	-1.22	-1.10	n/a	0.08	-0.09	-1.22
K1 Ib A0	0.00	-1.39	-0.77	n/a	n/a	n/a	n/a	n/a	-2.56	-2.88	n/a	-1.35	-1.57	-2.56
K1 Ib As	0.00	-0.25	0.13	n/a	n/a	n/a	n/a	n/a	-0.90	-0.83	n/a	-0.14	-0.31	-0.90
K1 Ib Al1	0.00	-0.30	0.00	n/a	n/a	n/a	n/a	n/a	-1.06	-0.96	n/a	-0.37	-0.59	-1.06
K1 Ib Al2	0.00	-0.33	-0.02	n/a	n/a	n/a	n/a	n/a	-1.14	-1.05	n/a	-0.22	-0.40	-1.14

window F (days 12-14)

	Persistence	Open loop (info-matched)	Open loop (structure-rich)*	LSTM (v1.0)	LSTM	PINN, single-stage	CL-PINN	PINN-θ	EKF	EKS	IEKS	Aug. EKF	Aug. EKS	Online EKF*
K0 Ie A0	0.00	0.63	1.00	-0.67	n/a	0.42	0.49	0.33†	0.12	0.12	-0.21	-0.36	-0.36	-0.13
K0 Ie As	0.00	0.63	1.00	n/a	n/a	n/a	n/a	n/a	0.09	0.09	n/a	-0.64	-0.64	-0.15
K0 Ie Al1	0.00	0.70	0.99	n/a	n/a	n/a	n/a	n/a	0.27	0.27	n/a	-0.31	-0.31	0.07
K0 Ie Al2	0.00	0.70	0.99	n/a	n/a	n/a	n/a	n/a	0.26	0.26	n/a	-0.37	-0.37	0.06
K0 Ic A0	0.00	0.52	0.61	n/a	n/a	n/a	0.46	n/a	-0.01	-0.01	-0.36	-1.07	-1.07	0.08
K0 Ic As	0.00	0.52	0.61	n/a	n/a	n/a	n/a	n/a	-0.04	-0.04	n/a	-1.49	-1.49	0.05
K0 Ic Al1	0.00	0.61	0.68	n/a	n/a	n/a	n/a	n/a	0.17	0.17	n/a	-0.98	-0.98	0.24
K0 Ic Al2	0.00	0.61	0.68	n/a	n/a	n/a	n/a	n/a	0.16	0.16	n/a	-1.03	-1.03	0.23
K0 Ib A0	0.00	0.53	0.63	n/a	n/a	n/a	n/a	n/a	0.29	0.29	n/a	-0.15	-0.15	0.16
K0 Ib As	0.00	0.53	0.63	n/a	n/a	n/a	n/a	n/a	0.25	0.25	n/a	-0.46	-0.46	0.14
K0 Ib Al1	0.00	0.62	0.70	n/a	n/a	n/a	n/a	n/a	0.41	0.41	n/a	-0.16	-0.16	0.31
K0 Ib Al2	0.00	0.62	0.70	n/a	n/a	n/a	n/a	n/a	0.40	0.40	n/a	-0.20	-0.20	0.30
K0 off-steady Ie A0	0.00	0.57	0.99	n/a	n/a	n/a	n/a	n/a	-0.01	-0.01	n/a	-0.55	-0.55	-0.28
K0 off-steady Ie As	0.00	0.68	0.92	n/a	n/a	n/a	n/a	n/a	0.18	0.18	n/a	-0.43	-0.43	-0.04
K0 off-steady Ie Al1	0.00	0.73	0.96	n/a	n/a	n/a	n/a	n/a	0.36	0.36	n/a	-0.12	-0.12	0.18
K.25 Ie A0	0.00	0.00	0.39	n/a	n/a	n/a	n/a	n/a	-0.15	-0.15	n/a	-0.31	-0.31	-0.26
K.25 Ie As	0.00	0.10	0.44	n/a	n/a	n/a	n/a	n/a	-0.01	-0.01	n/a	-0.67	-0.67	-0.10
K.25 Ie Al1	0.00	0.19	0.48	n/a	n/a	n/a	n/a	n/a	0.10	0.10	n/a	-0.47	-0.47	0.02
K.25 Ie Al2	0.00	0.18	0.47	n/a	n/a	n/a	n/a	n/a	0.08	0.08	n/a	-0.52	-0.52	-0.00
K.25 Ic A0	0.00	-0.03	0.23	n/a	n/a	n/a	n/a	n/a	-0.71	-0.71	n/a	-0.81	-0.81	-0.50
K.25 Ic As	0.00	0.08	0.31	n/a	n/a	n/a	n/a	n/a	-0.45	-0.45	n/a	-1.27	-1.27	-0.27
K.25 Ic Al1	0.00	0.17	0.37	n/a	n/a	n/a	n/a	n/a	-0.28	-0.28	n/a	-0.97	-0.97	-0.12
K.25 Ic Al2	0.00	0.16	0.35	n/a	n/a	n/a	n/a	n/a	-0.30	-0.30	n/a	-1.02	-1.02	-0.14
K.25 Ib A0	0.00	-0.02	0.23	n/a	n/a	n/a	n/a	n/a	-2.33	-2.33	n/a	-1.77	-1.77	-1.79
K.25 Ib As	0.00	0.08	0.32	n/a	n/a	n/a	n/a	n/a	-0.30	-0.30	n/a	-0.15	-0.15	-0.21
K.25 Ib Al1	0.00	0.18	0.37	n/a	n/a	n/a	n/a	n/a	-0.15	-0.15	n/a	-0.02	-0.02	-0.06
K.25 Ib Al2	0.00	0.16	0.36	n/a	n/a	n/a	n/a	n/a	-0.17	-0.17	n/a	-0.05	-0.05	-0.08
K.5 Ie A0	0.00	-0.71	-0.12	n/a	n/a	n/a	-1.26	n/a	-1.10	-1.10	-1.15	-0.27	-0.27	-1.34
K.5 Ie As	0.00	-0.13	0.26	n/a	n/a	n/a	n/a	n/a	-0.44	-0.44	n/a	-0.43	-0.43	-0.60
K.5 Ie Al1	0.00	-0.12	0.24	n/a	n/a	n/a	n/a	n/a	-0.43	-0.43	n/a	-0.38	-0.38	-0.58
K.5 Ie Al2	0.00	-0.17	0.21	n/a	n/a	n/a	n/a	n/a	-0.49	-0.49	n/a	-0.46	-0.46	-0.65
K.5 Ic A0	0.00	-0.76	-0.28	n/a	n/a	n/a	-1.57	n/a	-2.94	-2.94	-3.66	-2.28	-2.28	-2.32
K.5 Ic As	0.00	-0.16	0.17	n/a	-0.30	n/a	-0.67	n/a	-0.56	-0.56	-0.69	-0.48	-0.48	-0.46
K.5 Ic Al1	0.00	-0.15	0.15	n/a	n/a	n/a	-0.74	n/a	-0.54	-0.54	-0.66	-0.42	-0.42	-0.43
K.5 Ic Al2	0.00	-0.20	0.11	n/a	n/a	n/a	n/a	n/a	-0.61	-0.61	n/a	-0.48	-0.48	-0.50
K.5 Ib A0	0.00	-0.74	-0.26	n/a	n/a	n/a	n/a	n/a	-3.08	-3.08	n/a	-1.56	-1.56	-2.47
K.5 Ib As	0.00	-0.15	0.18	n/a	n/a	n/a	n/a	n/a	-1.62	-1.62	n/a	-1.28	-1.28	-1.22
K.5 Ib Al1	0.00	-0.14	0.16	n/a	n/a	n/a	n/a	n/a	-1.57	-1.57	n/a	-1.21	-1.21	-1.18
K.5 Ib Al2	0.00	-0.19	0.13	n/a	n/a	n/a	n/a	n/a	-1.68	-1.68	n/a	-1.23	-1.23	-1.28
K.75 Ie A0	0.00	-1.40	-0.57	n/a	n/a	n/a	n/a	n/a	-2.00	-2.00	n/a	-0.06	-0.06	-2.31
K.75 Ie As	0.00	-0.28	0.18	n/a	n/a	n/a	n/a	n/a	-0.73	-0.73	n/a	0.06	0.06	-0.91
K.75 Ie Al1	0.00	-0.33	0.11	n/a	n/a	n/a	n/a	n/a	-0.79	-0.79	n/a	0.06	0.06	-0.97
K.75 Ie Al2	0.00	-0.39	0.07	n/a	n/a	n/a	n/a	n/a	-0.89	-0.89	n/a	0.03	0.03	-1.08
K.75 Ic A0	0.00	-1.47	-0.73	n/a	n/a	n/a	n/a	n/a	-3.40	-3.40	n/a	-2.12	-2.12	-2.81
K.75 Ic As	0.00	-0.32	0.10	n/a	n/a	n/a	n/a	n/a	-1.27	-1.27	n/a	-1.11	-1.11	-0.96
K.75 Ic Al1	0.00	-0.37	0.03	n/a	n/a	n/a	n/a	n/a	-1.31	-1.31	n/a	-1.12	-1.12	-1.00
K.75 Ic Al2	0.00	-0.43	-0.02	n/a	n/a	n/a	n/a	n/a	-1.43	-1.43	n/a	-1.16	-1.16	-1.11
K.75 Ib A0	0.00	-1.45	-0.71	n/a	n/a	n/a	n/a	n/a	-2.91	-2.91	n/a	-0.78	-0.78	-2.71
K.75 Ib As	0.00	-0.31	0.11	n/a	n/a	n/a	n/a	n/a	-1.02	-1.02	n/a	-0.15	-0.15	-0.92
K.75 Ib Al1	0.00	-0.35	0.04	n/a	n/a	n/a	n/a	n/a	-1.06	-1.06	n/a	-0.11	-0.11	-0.95
K.75 Ib Al2	0.00	-0.42	-0.01	n/a	n/a	n/a	n/a	n/a	-1.16	-1.16	n/a	-0.08	-0.08	-1.06
K1 Ie A0	0.00	-2.08	-1.00	n/a	n/a	n/a	-2.98	-1.13	-2.71	-2.71	-2.40	-0.23	-0.23	-3.16
K1 Ie As	0.00	-0.40	0.13	n/a	n/a	n/a	n/a	n/a	-0.94	-0.94	n/a	0.19	0.19	-1.17
K1 Ie Al1	0.00	-0.48	0.03	n/a	n/a	n/a	n/a	n/a	-1.05	-1.05	n/a	0.22	0.22	-1.29
K1 Ie Al2	0.00	-0.54	-0.02	n/a	n/a	n/a	n/a	n/a	-1.16	-1.16	n/a	0.19	0.19	-1.41
K1 Ic A0	0.00	-2.18	-1.17	n/a	n/a	n/a	n/a	n/a	-3.03	-3.03	n/a	-0.51	-0.51	-3.00
K1 Ic As	0.00	-0.44	0.05	n/a	n/a	n/a	n/a	n/a	-0.79	-0.79	n/a	-0.32	-0.32	-0.78
K1 Ic Al1	0.00	-0.52	-0.05	n/a	n/a	n/a	n/a	n/a	-0.87	-0.87	n/a	-0.04	-0.04	-0.86
K1 Ic Al2	0.00	-0.59	-0.10	n/a	n/a	n/a	n/a	n/a	-0.97	-0.97	n/a	-0.16	-0.16	-0.96
K1 Ib A0	0.00	-2.16	-1.14	n/a	n/a	n/a	n/a	n/a	-3.28	-3.28	n/a	-1.75	-1.75	-3.20
K1 Ib As	0.00	-0.43	0.06	n/a	n/a	n/a	n/a	n/a	-0.86	-0.86	n/a	-0.04	-0.04	-0.83
K1 Ib Al1	0.00	-0.51	-0.04	n/a	n/a	n/a	n/a	n/a	-0.92	-0.92	n/a	-0.52	-0.52	-0.89
K1 Ib Al2	0.00	-0.58	-0.09	n/a	n/a	n/a	n/a	n/a	-1.02	-1.02	n/a	-0.33	-0.33	-0.99



Markers: dot = best on R0, triangle = best on F. * = more information, never a winner. † = PINN row with one or two seeds (labelled, Section 9). sigma = 0.10.